Problem 6: Class Attendance Tracker

TASK

A teacher records attendance for one student as an array of 1s (present) and 0s (absent), one value per day. The teacher wants to know how many days the student was present and the longest run of days the student attended without a break.

Implement a function `attendanceSummary(days)` that returns the number of present days and the length of the longest unbroken run of present days.

CONSTRAINTS

- 1 <= length of days <= 100
- Each value is 0 or 1

EXAMPLES

Sample 1: `days = [1, 1, 0, 1, 1, 1, 0, 1]`
Expected Output: Present: 6, Longest streak: 3

Sample 2: `days = [0, 0, 0]`
Expected Output: Present: 0, Longest streak: 0

SOLUTION IMPLEMENTATION

```
      def attendanceSummary(days): total_present = 0 max_streak = 0 current_streak = 0 for
day in days: if day == 1: total_present += 1 current_streak += 1 if current_streak >
max_streak: max_streak = current_streak else: current_streak = 0 return total_present,
max_streak
```

COMPLEXITY REQUIREMENTS

State the time and space complexity of your solution. Can both answers be found in a single pass?

Answer: Yes, both answers can be found in a single pass.
• **Time Complexity:** O(n) where n is the length of days.
• **Space Complexity:** O(1) using constant extra memory.